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For the attention of the EPO

European Patent Application No. 22748684.2
DEKA Products Limited Partnership

With reference to the International Preliminary Report on Patentability dated 14 December 2023 and the communication pursuant to Rules 161(1) and 162 EPC dated 9 February 2024, we file herewith amended page 1 of the description, newly-inserted page 1a and an amended set of claims to replace the claims currently on file.

We also enclose a marked-up copy of the amended pages on which the amendments are highlighted. For the Examining Division's ease of reference, the amended pages of the specification are show marked-up in relation to the specification of the application as filed rather than the application as amended on regional phase entry.

Amendments

Independent method claim 1 has been amended to specify that the method comprises "*securely transmitting the at least one rule database to the collection device*" rather than "*securely transmitting the at least one first filter to the collection device*". Basis for this amendment is provided by at least paragraph [0070] and Figure 6 of the application as filed.

Claim 1 has further been amended to specify that the steps of securely receiving at least one rule from an authorized actor, determining at least one first filter based on the at least one rule, updating at least one rule database with the at least one rule and securely transmitting the at least one rule database to the collection device are carried out by a processing device. Basis for these amendments is provided by at least Figure 6 and paragraph [0070] of the application as filed, which states that "*the method in FIG. 6 is written from the point of view of a processing device*".

Claim 1 has further been amended to specify that the method comprises the steps of: the collection device securely receiving the at least one rule database from the processing device;

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the collection device securely receiving and storing the sensor data associated with a pre-selected area associated with a location of the collection device; the collection device filtering the sensor data to determine filtered sensor data, the filtering based at least upon the at least one rule database; the collection device securely storing the filtered sensor data; and the collection device securely transmitting the filtered sensor data to the processing device. Basis for these amendments is provided by at least paragraph [0071] and Figure 7 of the application as filed.

Claim 1 has further been amended to specify that the method comprises "*securely receiving the filtered sensor data from the collection device*" rather than "*securely receiving the sensor data from the collection device, the sensor data being filtered by the at least one first filter*" for consistency with the preceding features of the claim.

Claim 1 has further been amended to specify that the steps of securely receiving the filtered sensor data from the collection device, determining the desired information by applying at least one second filter to the filtered sensor data, the at least one second filter being based at least upon the at least one rule database, and encrypting and transmitting the desired information to the authorized actor are carried out by the processing device. Basis for these amendments is provided by at least Figure 6 and paragraph [0070] of the application as filed, which, as mentioned above, states that "*the method in FIG. 6 is written from the point of view of a processing device*".

Claims 3 to 6, 12, 16, 17 and 23 to 114 have been deleted. The remaining claims and their dependencies have been renumbered accordingly.

Claim 18 (now claim 11) has further been amended to specify that the system executes on one or more processors. Basis for this amendment is provided by at least paragraph [0061] of the application as filed. Accordingly, claim 18 has further been amended to recite that it is a processor of the one or more processors on which the collection module is configured to be executed that is located in the autonomous vehicle.

Claim 18 has further been amended to specify that the system further comprises a processing module. Basis for this amendment is provided by at least paragraph [0061] of the application as filed.

The remaining amendments to claim 18 are those that have been made for consistency between claims 18 and 1.

US 2018/050800 A1 (D1) has been identified and briefly discussed in the background section of the description.

The Examining Division will note that the independent claims have not been placed in two-part form since this would require the features of the claims to be reordered in such a manner that the clarity of the independent claims, and, thus, the scope of the invention, would be compromised.

Unity

In the section regarding Item IV of the Written Opinion, the Examining Division objects that the claims relate to two distinct inventions.

As mentioned above, claims 3 to 6, 12, 16, 17 and 23 to 114 have now been deleted. Accordingly, only claims considered by the Examining Division to relate to the first invention remain in the claim set.

We therefore submit that the claims meet the requirements of unity.

Clarity

In the section regarding Item VIII of the Written Opinion, the Examining Division objects that independent claims 1, 11, 12 and 18 are not clear. In particular, the Examining Division comments that the terms "*securely receiving*", "*securely transmitting*" and "*authorized actor*" are broad and leave the reader in doubt as to the technical features to which they refer. We respectfully disagree with the Examining Division for at least the reasons outlined below.

Firstly, the Examining Division has not explicitly indicated what it is about the terms "*securely receiving*", "*securely transmitting*" and "*authorized actor*" that would leave the skilled person in doubt as to their meaning, other than to say that they are broad in scope. However, whether a feature is "*broad*" is subjective and cannot result in the feature lacking clarity *per se*. Instead, when assessing clarity, it is only relevant that the skilled person is able to determine what the scope of the claim is.

Merely for the sake of argument in the present response, we assume that the Examining Division considers the use of the word "*securely*" in the terms "*securely receiving*" and "*securely transmitting*" to result in a lack clarity, and, furthermore, considers the use of the word "*authorized*" in the term "*authorized actor*" to lack clarity.

Turning first to the use of the term "*authorized actor*", we submit that the skilled person would understand an authorized actor to be a person or entity that has been identified as being a person or entity from which the processing device is allowed to securely receive at least one rule. The skilled person would have no difficulty in identifying whether a prior art method or system or a third-party method or system disclosed or contained such a feature.

Regarding the use of the word "*securely*" in the terms "*securely receiving*" and "*securely transmitting*", we submit that the word "*securely*" would be understood by the skilled person to indicated that there are systems in place or characteristics of the method or system that prevent or inhibit unauthorised third-party access to the information (e.g. the rule database, sensor data and filtered sensor data) during its transmission between the processing device to the collection device. Again, the skilled person would have no difficulty in identifying whether a prior art method or system or a third-party method or system disclosed or contained such a feature.

We therefore submit that the claims meet the requirements of Article 84 EPC.

Patentability

In the sections regarding Items IV and V of the Written Opinion, the Examining Division objects that independent claim 1 lacks an inventive step over D1.

The Examining Division considers all of the features of claim 1 to be disclosed by D1, with the exception of the first, fourth and fifth steps being carried out <u>securely</u>, and the desired information being <u>encrypted</u> and transmitted to the authorized actor (underlined for emphasis). The Examining Division comments that these distinguishing features cannot be considered to involve an inventive step since they are a well-known design option that the skilled person would consider without inventive step, and cites paragraph [0061] as hinting at this feature since it refers to "*certain security measures in place to ensure that access is only allowed to authorized users*".

Although we disagree, as mentioned above, claim 1 has now been amended to include additional limitations that are highlighted in the attached marked-up set of claims.

The amendments to claim 1 involve, *inter alia*, specifying that the original steps of claim 1 are carried out by a processing device and specifying a number of additional steps of the method that are, in contrast, carried out by a collection device. The additional steps carried out by the collection device include:

- the collection device securely receiving the at least one rule database from the processing device;
- the collection device securely receiving and storing the sensor data associated with a pre-selected area associated with a location of the collection device;
- the collection device filtering the sensor data to determine filtered sensor data, the filtering based at least upon the at least one rule database;
- the collection device securely storing the filtered sensor data; and
- the collection device securely transmitting the filtered sensor data to the processing device.

In essence, the features of claim 1 as amended allow a rule database to be transmitted from a processing device to a collection device, the collection device to carry out a first filtering step of the sensor data based on the transmitted rule database, the filtered sensor data to be securely transmitted back to the processing device, and the processing device to then apply a second filter to the filtered sensor data to determine desired information that is provided to the authorized actor.

The method of claim 1 addresses security concerns in an efficient manner. In particular, by carrying out a first filtering step at the collection device (i.e. "*filtering the sensor data to determine filtered sensor data*") the amount of data that it is necessary to communicate from the collection device to the processing device is reduced. In addition, by carrying out a second filtering step at the processing device (i.e. "*applying at least one filter to the filtered sensor data*") rather than at the collection device, the data being transmitted from the collection device to the processing device remains only partially filtered, and, thus, less of a privacy concern. That is, the data being transmitted from the collection device to the processing device is less likely to have been filtered to the extent that third parties are able to deduce the overall criteria being used to filter the data.

The Examining Division will note that the additional steps carried out by the collection device that are now recited in independent claim 1 were previously recited by former independent claims 12 and 18. In section 3.1 of the Written Opinion, the Examining Division indicates that the features of former independent claim 1 are disclosed by paragraphs [0068], [0073] to [0078], [0090] and [0094] of D1, and indicates that the features of former independent claims 12 and 18 are disclosed by paragraphs [0090] to [0094] and Figures 1, 10 and 11 of D1.

D1 describes a police vehicle 10 equipped with a computer 12, a receiver 13 and a camera device 16. As shown in Figure 1 of D1, the computer 12 is in communication with the receiver 13 and is linked to a server 15 housed in a police station 14 via a communication network 18. The computer 12 is configured to detect objects of interest. D1 describes that law enforcement agencies often issue "*be on look out*" (BOLO) alerts to bring to the attention of law enforcement officers key information regarding an occurrence or activity of high importance. Upon receipt of such alerts by the receiver 13, the computer 12 activates the camera device 16 to start collecting

information and processing the captured image data to determine whether the specific content identified in the alert is present in the captured image data. In particular, when an object enters the scene, the computer 12 performs analytics on the captured image data using an analytics engine that references a database that is onboard the vehicle 10 or remotely accessed via the communication network 18. Once the analytics engine detects a match or near match of the designated content in the captured image data, the analytics engine proceeds to another step of further analyzing the data containing the designated content to detect for the presence of one or more designated details or attributes of or associated with the designated content.

In the abovementioned example of D1, all of the analysis of the image data is done on the analytics engine located on the computer 12 in the vehicle 10. Accordingly, in the abovementioned example of D1, there is no disclosure of the collection device securely storing the filtered sensor data, the collection device securely transmitting the filtered sensor data to the processing device, the processing device securely receiving the filtered sensor data from the collection device, and the processing device determining the desired information by applying at least one second filter to the filtered sensor data, the at least one second filter being based at least upon the at least one rule database. There is also no disclosure in the abovementioned example of D1 of the processing device encrypting and transmitting the desired information to the authorized actor.

It is noted that paragraph [0077] of D1 does describe that "*in some embodiments, the analytics engine is configured to detect unknown objects (e.g. a modified vehicle)*" and that "*this data can be locally stored for later upload or immediately transmitted to another location (e.g. to server 15) for verification and/or classification to aid in the training of detection of objects by the detection engine*". However, this does not amount to disclosure of the invention. Instead, this paragraph indicates that, if the analytics engine is unable to carry out filtering because the object is unknown, the step of verification and/or classification is instead offloaded to the server 15. This goes completely against the fundamental teaching of the invention, which, as mentioned above, essentially involves a collection device carrying out a first filtering step of the sensor data based on the transmitted rule database, the filtered sensor data being securely transmitted back to the processing device and a processing device then applying a second filter to the filtered sensor data to determine desired information that is provided to the authorized actor.

Paragraph [0077] also recites that "*in some embodiments, the analytics for recognition and detection of the designated content is distributed among the vehicle 10 computer 12 and one or more remote computers (e.g. the server 15 in the police station 14)*". However, this is merely a general disclosure encompassing the abovementioned scenario in which filtering is carried out in the vehicle 10 if filtering in the vehicle 10 is possible, and in the server 15 if filtering in the vehicle 10 is not possible. This does not amount to disclosure of analysis being carried out on the same data in two different stages (i.e. firstly at a collection module and secondly at a processing module).

As suggested above, the objective technical problem of the invention may be regarded as how to address security concerns in an efficient manner. However, when faced with this objective technical problem, it would not be possible for the skilled person to arrive at the claimed invention. In particular, D1 fails to disclose the distinguishing features of the claimed invention, and teaches that verification and/or classification should be carried out at the server 15 only if the analytics engine located on the computer 12 in the vehicle 10 is unable to carry out the filtering. This goes completely against the teaching of the invention and is done for a completely different reason than the invention (i.e. simply to provide redundancy rather than to address security concerns in an efficient manner). There is simply nothing in D1 that would lead the skilled person to modify D1 to go against the teaching of D1 and incorporate the distinguishing

features of the invention, and any assertion that such modifications would be obvious is based on hindsight, which is not permitted under European practice.

For at least these reasons, claim 1 is both novel and inventive over the cited prior art. The same arguments apply, mutatis mutandis, to corresponding independent claim 18 against which an inventive step objection has also been raised. Claim 18 is therefore both novel and inventive over the cited prior art. The remaining claims are both novel and inventive at least by virtue of their dependency.

Further comments

Any amendments made by way of the present letter, and the observations contained herein, are made solely for the purposes of the prosecution of this European patent application and without prejudice to the applicant in other jurisdictions.

It should be noted that insofar as any subject-matter might otherwise be regarded as having been abandoned or effectively disclaimed by virtue of amendments mentioned above and/or incorporated in the new pages submitted with this letter, the applicant wishes to reserve the right to restore such subject matter and/or file a divisional application in respect thereof.

We consider this response to be a full response to the communication. However, if there are any remaining objections which may be usefully dealt with by telephone, the Examiner is requested to contact us on the email address provided with this online filing to arrange a time to discuss. We hereby take the precaution of requesting oral proceedings in the event that the Examiner is minded to refuse the application, but we would prefer to continue prosecution of this application by written correspondence or by telephone interview.

Yours faithfully

[SIGNED ELECTRONICALLY]

Jack Whitfield
Authorised Representative
HASELTINE LAKE KEMPNER LLP